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Spring Secuirty

With Boot

Study Guide

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Spring Security with Boot

Lab28: Spring Security with Boot

Working Steps:

1. Copy Lab16 as Lab28.

2. Remove the following.

*BooksList.jsp

*Book.java

*BooksController.java

3. Update pom.xml.

- Just change the Names

4. Build once.

-It shows index.jsp if everything is fine.

5. Add the following JSP's

index.jsp

mymenu.jsp

myheader.jsp

myfooter.jsp

invalidAccess.jsp

viewBooks.jsp

addBook.jsp

editBook.jsp

deleteBook.jsp

placeOrder.jsp

6. Add the following Controller

UserController.java

7. Deploy and Run

8. Check the following URL's

<http://localhost:12345/login>

<http://localhost:12345/logout>

<http://localhost:12345/>

<http://localhost:12345/viewBooks>

<http://localhost:12345/addBook>

<http://localhost:12345/editBook>

<http://localhost:12345/deleteBook>

<http://localhost:12345/placeOrder>



Steps to Add Spring Security

9. Add Maven Dependency in pom.xml

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-security</artifactId>
</dependency>
```

Note : we will have now following 8 dependencies

- spring-boot-starter-security ***
- spring-boot-starter-web
- tomcat-embed-jasper
- jstl
- standard
- bootstrap
- jquery
- spring-boot-devtools

10.Bean Configuration class has to extend **WebSecurityConfigurerAdapter**

```
@SpringBootApplication
public class JLCWebConfig extends WebSecurityConfigurerAdapter implements
WebMvcConfigurer{

}
```

11.Authetication

- Configure a Method with AuthenticationManagerBuilder.
- AuthenticationManagerBuilder is used to create the required AuthenticationManager
- void configureGlobal(AuthenticationManagerBuilder auth){
 //Setup AuthenticationManager
}
- There are 3 AuthenticationManagers.
 - a) InMemoryAuthenticationManager
 - b) JdbcAuthenticationManager
 - c) LdapAuthenticationManager
- AuthenticationManagerBuilder has the following 3 methods to create the required AuthenticationManager
 - a) auth.inMemoryAuthentication()
 - b) auth.jdbcAuthentication()
 - c) auth.ldapAuthentication()



- We need to Configure Usernames, Passwords and Roles of the Users with AuthenticationManagers with the following method.

```
void configureGlobal(AuthenticationManagerBuilder auth){  
  
}
```

- In this Lab, We are using InMemoryAuthenticationManager

```
public void configureGlobal(AuthenticationManagerBuilder auth) {  
    auth.inMemoryAuthentication()  
        .withUser("srinivas")  
        .password("{noop}srinivas")  
        .roles("ADMIN");  
}
```

12.Authorization

Decide the Operations Allowed for Each Role.

Customer	viewBooks , placeOrder
STOREKEEPER	viewBooks , addBook, updateBook
Admin	viewBooks , addBook, updateBook , deleteBook

Now Override the following method

```
@Override  
protected void configure(HttpSecurity http) throws Exception {  
    http.authorizeRequests()  
        .antMatchers("/deleteBook**").access("hasRole('ROLE_ADMIN')")  
    }  
}
```

13.Add invalidAccess.jsp

14.Deploy and Run

Lab28: Files Required

1. index.jsp	2. myheader.jsp
3. myfooter.jsp	4. mymenu.jsp
5. viewBooks.jsp	6. addBook.jsp
7. editBook.jsp	8. deleteBook.jsp
9. placeOrder.jsp	10. invalidAccess.jsp
11. UserController.java	12. JLCWebConfig.java
13. StartMyBootApplication.java	14. application.properties
15. pom.xml	



1. index.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>

<html>  <head>
    <title>JLC Bookstore</title>
    <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<div class="card">
<div class="card-body">
<div class="container" align="center">
<h2>
<a href='<c:url value="viewBooks"/>' class="btn btn-success btn-lg"> View Books</a>
<br/><br/>
<a href='<c:url value="addBook"/>' class="btn btn-success btn-lg"> Add Book </a>
<br/><br/>
<a href='<c:url value="editBook"/>' class="btn btn-success btn-lg"> Edit Book </a>
<br/><br/>
<a href='<c:url value="deleteBook"/>' class="btn btn-success btn-lg"> Delete Book </a>
<br/><br/>
<a href='<c:url value="placeOrder"/>' class="btn btn-success btn-lg"> Place Order </a>
</h2>
</div> </div>
</div>
<c:import url="myfooter.jsp"/>
</body> </html>
```

2. myheader.jsp

```
<div class="card-header">
<h2 class="text-center">Welcome to JLC Bookstore</h2>
<h4 class="text-center"> Best Books and Best Videos</h4>
</div>
```

3. myfooter.jsp

```
<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>
```



4. mymenu.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<div class="container" align="right">
<h2>
| <a href="<c:url value='/'/>"> Home</a>
| Welcome : ${pageContext.request.userPrincipal.name}
| <a href="<c:url value='logout'/>"> Logout </a>
</h2>
</div>
```

5. viewBooks.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<html>
<head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
```

```
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
```

```
<div class="card">
  <div class="card-body">
    <c:import url="mymenu.jsp"/>
```

```
<div class="container">
  <hr>
  <h2> Display List of books here</h2>
  <h2> Display List of books here</h2>
  <h2> Display List of books here</h2>
  <h2> Display List of books here</h2>
  <h2> Display List of books here</h2>
</div>
</div>
</div>
  <c:import url="myfooter.jsp"/>
</body>
</html>
```



6. addBook.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>

<html> <head>
    <title>JLC Bookstore</title>
    <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>

<div class="card">
    <div class="card-body">
        <c:import url="mymenu.jsp"/>

<div class="container">
    <hr>
<h2> Add Book Form Here</h2>
<h2> Add Book Form Here</h2>
<h2> Add Book Form Here</h2>
<h2> Add Book Form Here</h2>
<h2> Add Book Form Here</h2>
</div>
</div>
</div>

    <c:import url="myfooter.jsp"/>
</body> </html>
```

7. editBook.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>

<html> <head>
    <title>JLC Bookstore</title>
    <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
```



```
<div class="card">
  <div class="card-body">
    <c:import url="mymenu.jsp"/>

    <div class="container">
      <hr>
      <h2> Edit Book Form Here</h2>
      <h2> Edit Book Form Here</h2>
      <h2> Edit Book Form Here</h2>
      <h2> Edit Book Form Here</h2>
      <h2> Edit Book Form Here</h2>
    </div>
  </div>
</div>

<c:import url="myfooter.jsp"/>
</body> </html>
```

8. deleteBook.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>

<html>
<head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<div class="card">
  <div class="card-body">
    <c:import url="mymenu.jsp"/>

    <div class="container">
      <hr>

      <h2> Delete Book Form Here</h2>
      <h2> Delete Book Form Here</h2>
      <h2> Delete Book Form Here</h2>
      <h2> Delete Book Form Here</h2>
      <h2> Delete Book Form Here</h2>
    </div>
```



```
</div>
</div>
<c:import url="myfooter.jsp"/>
</body> </html>
```

9. placeOrder.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>

<html> <head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<div class="card">
  <div class="card-body">
    <c:import url="mymenu.jsp"/>

<div class="container">
  <hr>
<h2> Your Order Placed</h2>
<h2> Your Order Placed</h2>
<h2> Your Order Placed</h2>
<h2> Your Order Placed</h2>
</div>
</div>
</div>
  <c:import url="myfooter.jsp"/>
</body> </html>
```

10. invalidAccess.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>

<html> <head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<div class="card">
  <div class="card-body">
```



```
<c:import url="mymenu.jsp"/>

<div class="container">
<hr>
<h2 align="center">You are not authorized to access this page</h2>
</div>
</div>
</div>
<c:import url="myfooter.jsp"/>
</body> </html>
```

11. UserController.java

```
package com.coursecube.springboot;

import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @company : CourseCube
 * @website : www.coursecube.com
 */
@Controller
public class UserController {

    @GetMapping("/")
    public String showIndexPage(Model model) {
        System.out.println("** showIndexPage **");

        return "index";
    }

    @GetMapping("/viewBooks")
    public String showViewBooksPage(Model model) {
        System.out.println("** showViewBooksPage **");

        return "viewBooks";
    }

    @GetMapping("/addBook")
    public String showAddBookPage(Model model) {
        System.out.println("** showAddBookPage **");
        return "addBook";
    }
}
```



@GetMapping("/editBook")

```
public String showEditBookPage(Model model) {  
    System.out.println("** showEditBookPage **");
```

```
    return "editBook";  
}
```

@GetMapping("/deleteBook")

```
public String showDeleteBookPage(Model model) {  
    System.out.println("** showDeleteBookPage **");
```

```
    return "deleteBook";  
}
```

@GetMapping("/placeOrder")

```
public String showPlaceOrderPage(Model model) {  
    System.out.println("** showPlaceOrderPage **");
```

```
    return "placeOrder";  
}  
}
```

12. ILCWebConfig.java

```
package com.coursecube.springboot;
```

```
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.boot.autoconfigure.SpringBootApplication;  
import org.springframework.context.annotation.Bean;  
import  
org.springframework.security.config.annotation.authentication.builders.AuthenticationMa  
nagerBuilder;  
import org.springframework.security.config.annotation.web.builders.HttpSecurity;  
import  
org.springframework.security.config.annotation.web.configuration.WebSecurityConfigurer  
Adapter;  
import org.springframework.web.servlet.ViewResolver;  
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;  
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;  
import org.springframework.web.servlet.view.InternalResourceViewResolver;  
import org.springframework.web.servlet.view.JstlView;  
/*  
 * @Author : Srinivas Dande  
 * @company : CourseCube  
 * @website : www.coursecube.com  
 */
```



@SpringBootApplication

public class JLCWebConfig extends WebSecurityConfigurerAdapter implements
WebMvcConfigurer{

@Bean

```
public ViewResolver viewResolver() {  
    InternalResourceViewResolver resolver = new InternalResourceViewResolver();  
    resolver.setPrefix("WEB-INF/myjsps/");  
    resolver.setSuffix(".jsp");  
    resolver.setViewClass(JstlView.class);  
    return resolver;  
}
```

@Override

```
public void addResourceHandlers(ResourceHandlerRegistry registry) {  
    registry.addResourceHandler("/webjars/**")  
        .addResourceLocations("classpath:/META-INF/resources/webjars/");  
}
```

@Autowired

```
public void configureGlobal(AuthenticationManagerBuilder auth) throws Exception  
{  
    auth.inMemoryAuthentication()  
        .withUser("srinivas").password("{noop}srinivas").roles("ADMIN");  
    auth.inMemoryAuthentication()  
        .withUser("vas").password("{noop}vas").roles("STOREKEEPER");  
    auth.inMemoryAuthentication()  
        .withUser("sri").password("{noop}sri").roles("CUSTOMER");  
    auth.inMemoryAuthentication()  
        .withUser("sd").password("{noop}sd").roles("CUSTOMER");  
    auth.inMemoryAuthentication()  
        .withUser("super").password("{noop}super").roles("CUSTOMER","ADMIN");  
}
```

@Override

```
protected void configure(HttpSecurity http) throws Exception {  
    http.authorizeRequests()  
        .antMatchers("/deleteBook***").access("hasRole('ROLE_ADMIN')")  
        .antMatchers("/addBook***").access("hasAnyRole('ROLE_ADMIN','ROLE_STOREKEEPER')")  
        )  
        .antMatchers("/editBook***").access("hasAnyRole('ROLE_ADMIN','ROLE_STOREKEEPER')")  
        )  
        .antMatchers("/placeOrder***").access("hasAnyRole('ROLE_CUSTOMER')")  
        .and()  
}
```



```
.formLogin()  
.and()  
.exceptionHandling().accessDeniedPage("/WEB-INF/myjsps/invalidAccess.jsp");  
}  
}
```

13. StartMyBootApplication.java

```
package com.coursecube.springboot;  
  
import org.springframework.boot.SpringApplication;  
import org.springframework.boot.autoconfigure.SpringBootApplication;  
/*  
 * @Author : Srinivas Dande  
 * @company : CourseCube  
 * @website : www.coursecube.com  
 */  
@SpringBootApplication  
public class StartMyBootApplication{  
    public static void main(String[] args) {  
        System.out.println("My Boot App - main - begin");  
        SpringApplication.run(StartMyBootApplication.class, args);  
        System.out.println("My Boot App - main - end");  
    }  
}
```

14. application.properties

```
server.port=12345  
  
logging.level.root=INFO  
logging.pattern.console=%-5level %logger{36} - %msg%n
```

15. pom.xml

```
<project...>  
<modelVersion>4.0.0</modelVersion>  
<groupId>com.coursecube.springboot</groupId>  
<artifactId>Lab28</artifactId>  
<version>1.0</version>  
<name>MyLab28</name>  
<packaging>jar</packaging>  
<url>http://www.coursecube.com</url>  
  
<properties>  
<maven-jar-plugin.version>3.1.1</maven-jar-plugin.version>  
</properties>
```



```
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-security</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>
<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>MyLab28</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```

Spring Security with Boot

Improvements required in Lab28:

- A) Display either login/Register or logout
- B) Display the Links according to Role.
- C) Customize Login Page and Error Messages.
- D) Customize Logout Page
- E) Display the Role after the username.
 - Display the following for Authenticated Users
Home Welcome : sri [ROLE_CUSTOMER] Logout
 - Display the following for Un-Authenticated Users
Home Login Register

Lab29: Working Steps:

- 1) Copy Lab28 and paste as Lab29
- 2) Change the Names in pom.xml

```
<groupId>com.coursecube.springboot</groupId>
<artifactId>Lab29</artifactId>
<version>1.0</version>
<name>MyLab29</name>
```



Steps for Task A: Display either login/Register or logout

3) Add the Maven Dependency for spring-security-taglibs

```
<dependency>  
<groupId>org.springframework.security</groupId>  
<artifactId>spring-security-taglibs</artifactId>  
</dependency>
```

4) Use the following taglib directive in the required jsp.(mymenu.jsp)

```
<%@ taglib prefix="sec" uri="http://www.springframework.org/security/tags"%>
```

5) Update mymenu.jsp

```
<sec:authorize access="! isAuthenticated()">  
| <a href='<c:url value="login"/>'>Login</a>  
| <a href='<c:url value="register"/>'>Register</a>  
</sec:authorize>
```

```
<sec:authorize access="isAuthenticated()">  
<a href='<c:url value="logout"/>'>Logout</a>  
</sec:authorize>
```

Steps for Task B: Display the Links according to Role.

6) Update index.jsp

```
<sec:authorize access="hasRole('ADMIN')">  
<a href='<c:url value="deleteBook"/>'> Delete Book </a>  
</sec:authorize>
```

```
<sec:authorize access="hasRole('CUSTOMER')">  
<a href='<c:url value="placeOrder"/>'> Place Order </a>  
</sec:authorize>
```

Steps for Task C: Customize Login Page and Error Messages.

7) Design the mylogin.jsp

8) Update HttpSecurity with login configuration.

```
.formLogin()  
.loginPage("/login")  
.failureUrl("/login?error")  
.usernameParameter("myusername")  
.passwordParameter("mypassword")
```



Types of Authentication:

- 1) Basic Authentication
- 2) Form based Authentication (used in this Lab)
- 3) Digest Authentication
- 4) Client Certificate Authentication

9) Add the method in the controller to handle request for /login

```
@GetMapping("/login")
public String loginPage(...) {
    .....
    //Get the Cart Data if any and store in Session
    //Get the Last Accessed Time from DB.

    return "mylogin";
}
```

10) Test the Application

A .http://localhost:56789/login

No Parameters going to server with names error or logout
** Nothing display in login Page

B.http://localhost:56789/login?error

error Parameter going to server

```
<c:if test="${param.error != null}">
    Invalid username and password.
</c:if>
```

C.http://localhost:56789/login?logout

logout Parameter going to server

```
<c:if test="${param.logout != null}">
    You have logged out successfully.
</c:if>
```

Steps for Task D: Customize Logout Page

11) Update HttpSecurity with logout configuration.

```
.logout()
.invalidateHttpSession(true)
.logoutSuccessUrl("/login?logout")
```



12) Add the method in the controller to handle request for /logout

```
@GetMapping("/logout")
public String logoutPage(.....) {
    //Store the Cart Data if any into the database
    Tasks At Logout time
}
```

Steps for Task E : Display the Role after the username.

13) Write the method in controller to get the roles of the current Authenticated user.

```
List<String> getUserRoles() {
    ..
}
```

14) Get the Roles list from getUserRoles() method and store that as attribute.

```
List<String> myroles=getUserRoles();
session.addAttribute("MyRoles", myroles);
```

15) Display the attribute in mymenu.jsp

```
${MyRoles}
```

Lab29: Files Required

1. index.jsp	Updated in Lab29
2. myheader.jsp	Same as Lab28
3. myfooter.jsp	Same as Lab28
4. mymenu.jsp	Updated in Lab29
5. viewBooks.jsp	Same as Lab28
6. addBook.jsp	Same as Lab28
7. editBook.jsp	Same as Lab28
8. deleteBook.jsp	Same as Lab28
9. placeOrder.jsp	Same as Lab28
10. invalidAccess.jsp	Same as Lab28
11. mylogin.jsp	Newly Added in Lab29
12. UserController.java	Updated in Lab29
13. JLCWebConfig.java	Updated in Lab29
14. StartMyBootApplication.java	Same as Lab28
15. application.properties	Same as Lab28
16. pom.xml	Updated in Lab29



1. index.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<%@ taglib prefix="sec" uri="http://www.springframework.org/security/tags" %>

<html> <head>
    <title>JLC Bookstore</title>
    <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>

<div class="card">
    <div class="card-body">
        <c:import url="mymenu.jsp"/>
        <div class="container" align="center">
<h2>

<a href='<c:url value="viewBooks"/>'> View Books</a>

<sec:authorize access="hasRole('ADMIN') or hasRole('STOREKEEPER')">
<br/><br/>
<a href="addBook"> Add Book</a>
<br/><br/>
<a href="editBook"> Edit Book</a>
</sec:authorize>

<sec:authorize access="hasRole('ADMIN')">
<br/><br/>
<a href="deleteBook"> Delete Book</a>
</sec:authorize>

<sec:authorize access="hasRole('CUSTOMER')">
<br/><br/>
<a href="placeOrder"> Place Order</a>
</sec:authorize>
</h2>
</div>
</div>
</div>
    <c:import url="myfooter.jsp"/>
</body> </html>
```



2. **myheader.jsp**

3. **myfooter.jsp**

4. **mymenu.jsp**

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<%@ taglib prefix="sec" uri="http://www.springframework.org/security/tags" %>
```

```
<div class="container" align="right">
```

```
<h2>
```

```
<a href="/">Home</a>
```

```
| Welcome : ${pageContext.request.userPrincipal.name} ${MyRoles}
```

```
<sec:authorize access="! isAuthenticated()">
```

```
| <a href="login">Login</a>
```

```
| <a href="showRegister">Register</a>
```

```
</sec:authorize>
```

```
<sec:authorize access="isAuthenticated()">
```

```
<a href="logout"> Logout</a>
```

```
</sec:authorize>
```

```
</h2>
```

```
</div>
```

5. **viewBooks.jsp**

6. **addBook.jsp**

7. **editBook.jsp**

8. **deleteBook.jsp**

9. **placeOrder.jsp**

10. **invalidAccess.jsp**

11. **mylogin.jsp**

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<html> <head>
```

```
<title>JLC Bookstore</title>
```

```
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
```

```
</head>
```

```
<body>
```

```
<div class="card">
```

```
<c:import url="myheader.jsp"/>
```

```
</div>
```

```
<div class="card">
```

```
<div class="card-body" align="center">
```



```
<div class="container-md">
<h2 align="center">
<c:if test="${param.error != null}">
<font color="red" size=5> Invalid username and password. </font>
</c:if>
<c:if test="${param.logout != null}">
<font color="red" size=5> You have logged out successfully. </font>
</c:if>
</h2>
</div>

<div class="container form-group">
<form name='loginForm' action="<c:url value='/login' />" method='POST'>
<table class="table">
<tr>
<td align="center" colspan="2">
<h2>User Login Form</h2>
</td> </tr>
<tr>
<td><label for="myusername"> Username </label></td>
<td><input type='text' name='myusername' id="myusername" class="form-control form-control-lg">
</td> </tr>
<tr>
<td>Password</td>
<td><input type='password' name='mypassword' class="form-control form-control-lg" />
</td></tr>
<tr>
<td align="center" colspan="2">
<input type=hidden name="${_csrf.parameterName}" value="${_csrf.token}" />
<input type="submit" value="Login Now" class="btn btn-primary btn-lg">
</td> </tr>
</table>
</form>
</div>
<div class="container">
<h2> <b> <a href='<c:url value="showRegister" />'> New User Register Here </a> </b> </h2>
</div>
</div> </div>
<c:import url="myfooter.jsp" />
</body>
</html>
```



12. UserController.java

```
package com.coursecube.springboot;
```

```
import java.util.ArrayList;  
import java.util.Collection;  
import java.util.List;
```

```
import javax.servlet.http.HttpServletRequest;  
import javax.servlet.http.HttpServletResponse;  
import javax.servlet.http.HttpSession;
```

```
import org.springframework.security.core.Authentication;  
import org.springframework.security.core.GrantedAuthority;  
import org.springframework.security.core.context.SecurityContextHolder;  
import  
org.springframework.security.web.authentication.logout.SecurityContextLogoutHandler;  
import org.springframework.stereotype.Controller;  
import org.springframework.ui.Model;  
import org.springframework.web.bind.annotation.GetMapping;  
import org.springframework.web.bind.annotation.RequestParam;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company : CourseCube  
 * @Website : www.coursecube.com  
 */
```

```
@Controller
```

```
public class UserController {
```

```
@GetMapping("/")
```

```
public String showIndexPage(Model model,HttpSession session) {  
System.out.println("** showIndexPage **");
```

```
List<String> myroles=getUserRoles();  
session.setAttribute("MyRoles", myroles);
```

```
return "index";  
}
```

```
@GetMapping("/viewBooks")
```

```
public String showViewBooksPage(Model model) {  
System.out.println("** showViewBooksPage **");
```

```
return "viewBooks";  
}
```



@GetMapping("/addBook")

```
public String showAddBookPage(Model model) {  
    System.out.println("** showAddBookPage **");
```

```
    return "addBook";  
}
```

@GetMapping("/editBook")

```
public String showEditBookPage(Model model) {  
    System.out.println("** showEditBookPage **");
```

```
    return "editBook";  
}
```

@GetMapping("/deleteBook")

```
public String showDeleteBookPage(Model model) {  
    System.out.println("** showDeleteBookPage **");
```

```
    return "deleteBook";  
}
```

@GetMapping("/placeOrder")

```
public String showPlaceOrderPage(Model model) {  
    System.out.println("** showPlaceOrderPage **");
```

```
    return "placeOrder";  
}
```

@GetMapping("/login")

```
public String loginPage(  
    @RequestParam(value = "error", required = false) String error,  
    @RequestParam(value = "logout", required = false) String logout, HttpSession session) {  
    System.out.println("** loginPage **");
```

```
    return "mylogin";  
  
}
```



```
@GetMapping("/logout")
```

```
public String logoutPage(HttpServletRequest request, HttpServletResponse  
response) {
```

```
System.out.println("** logoutPage **");
```

```
Authentication auth = SecurityContextHolder.getContext().getAuthentication();
```

```
if (auth != null) {
```

```
new SecurityContextLogoutHandler().logout(request, response, auth);
```

```
}
```

```
return "redirect:/login?logout";
```

```
}
```

```
private List<String> getUserRoles() {
```

```
List<String> myroles = new ArrayList<String>();
```

```
Authentication auth = SecurityContextHolder.getContext().getAuthentication();
```

```
if (auth != null) {
```

```
Collection<GrantedAuthority> col = (Collection<GrantedAuthority>) auth.getAuthorities();
```

```
for (GrantedAuthority gauth : col)
```

```
myroles.add(gauth.getAuthority());
```

```
}
```

```
return myroles;
```

```
}
```

```
}
```

13. JLCWebConfig.java

```
package com.coursecube.springboot;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
import org.springframework.context.annotation.Bean;
```

```
import
```

```
org.springframework.security.config.annotation.authentication.builders.AuthenticationManager  
Builder;
```

```
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
```

```
import
```

```
org.springframework.security.config.annotation.web.configuration.WebSecurityConfigurerAdapt  
er;
```

```
import org.springframework.web.servlet.ViewResolver;
```

```
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
```

```
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
```

```
import org.springframework.web.servlet.view.InternalResourceViewResolver;
```

```
import org.springframework.web.servlet.view.JstlView;
```



```
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
@SpringBootApplication
public class JLCWebConfig extends WebSecurityConfigurerAdapter implements
WebMvcConfigurer{

    @Bean
    public ViewResolver viewResolver() {
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();
        resolver.setPrefix("WEB-INF/myjsps/");
        resolver.setSuffix(".jsp");
        resolver.setViewClass(JstlView.class);
        return resolver;
    }

    @Override
    public void addResourceHandlers(ResourceHandlerRegistry registry) {
        registry.addResourceHandler("/webjars/**")
        .addResourceLocations("classpath:/META-INF/resources/webjars/");
    }

    @Autowired
    public void configureGlobal(AuthenticationManagerBuilder auth) throws Exception {

        auth.inMemoryAuthentication().withUser("srinivas").password("{noop}srinivas").roles("ADMIN
");
        auth.inMemoryAuthentication().withUser("vas").password("{noop}vas").roles("STOREKEEPER"
);
        auth.inMemoryAuthentication().withUser("sri").password("{noop}sri").roles("CUSTOMER");
        auth.inMemoryAuthentication().withUser("sd").password("{noop}sd").roles("CUSTOMER");
        auth.inMemoryAuthentication().withUser("super").password("{noop}super").roles("CUSTOMER
","ADMIN");
    }

    @Override
    protected void configure(HttpSecurity http) throws Exception {
        http.authorizeRequests()
        .antMatchers("/deleteBook***").access("hasRole('ROLE_ADMIN')")
        .antMatchers("/addBook***").access("hasAnyRole('ROLE_ADMIN','ROLE_STOREKEEPER')")
        .antMatchers("/editBook***").access("hasAnyRole('ROLE_ADMIN','ROLE_STOREKEEPER')")
    }
}
```



```
.antMatchers("/placeOrder**").access("hasAnyRole('ROLE_CUSTOMER')")
.and()
.formLogin()
.loginPage("/login")
.failureUrl("/login?error")
.usernameParameter("myusername")
.passwordParameter("mypassword")
.and()
.logout()
.invalidateHttpSession(true)
.logoutSuccessUrl("/login?logout")
.and()
.exceptionHandling().accessDeniedPage("/WEB-INF/myjsps/invalidAccess.jsp");
}
}
```

14. StartMyBootApplication.java

15. application.properties

16. pom.xml

```
<project...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.coursecube.springboot</groupId>
<artifactId>Lab29</artifactId>
<version>1.0</version>
<name>MyLab29</name>
<packaging>jar</packaging>
<url>http://www.coursecube.com</url>

<properties>
<maven-jar-plugin.version>3.1.1</maven-jar-plugin.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-security</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.springframework.security</groupId>
<artifactId>spring-security-taglibs</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>
```



```
<build>  
<finalName>MyLab29</finalName>  
<plugins>  
<plugin>  
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-maven-plugin</artifactId>  
</plugin>  
</plugins>  
</build>  
</project>
```



Lab30 : Spring Security with Boot

Improvements required in Lab28:

1. Display either login/Register or logout
2. Display the Role after the username.
 - Display the following for Authenticated Users
Home Welcome : sri [ROLE_CUSTOMER] Logout
 - Display the following for Un-Authenticated Users
Home Welcome : Login Register
3. Customize Login Page and Error Messages.
4. Customize Logout Page
5. Use the JDBCAuthenticationManager.
6. Implement Registration and Use PasswordEnoders
7. Display the Links according to Role.

Lab16: Working Steps:

- 1) Copy **Lab28** and paste as **Lab30**
- 2) Change the Names in pom.xml

Steps for Task 1: Display either login/Register or logout

- 3) Add the Maven Dependency for spring-security-taglibs

```
<dependency>
<groupId>org.springframework.security</groupId>
<artifactId>spring-security-taglibs</artifactId>
</dependency>
```

- 4) Use the following taglib directive in the required jsp.(mymenu.jsp)

```
<%@ taglib prefix="sec" uri="http://www.springframework.org/security/tags"%>
```



5) Update mymenu.jsp

```
<sec:authorize access="! isAuthenticated()">
| <a href='<c:url value="login"/>'>Login</a>
| <a href='<c:url value="register"/>'>Register</a>
</sec:authorize>
```

```
<sec:authorize access="isAuthenticated()">
<a href="<c:url value='logout'/>"> Logout</a>
</sec:authorize>
```

Steps for Task 2: Display the Role after the username.

6) Write the method to get the roles of the current Authenticated user.

```
List<String> getUserRoles() {
..
}
```

7) Get the Roles list from getUserRoles() method and store that as attribute.

```
List<String> myroles=getUserRoles();
session.addAttribute("MyRoles", myroles);
```

8) Display the attribute in mymenu.jsp

```
${MyRoles}
```

Steps for Task 3: Customize Login Page and Error Messages.

9) Design the mylogin.jsp

10) Update HttpSecurity with login configuration.

```
.formLogin()
.loginPage("/login")
.failureUrl("/login?error")
.usernameParameter("myusername")
.passwordParameter("mypassword")
```

Types of Authentication:

- 1) Basic Authentication
- 2) Form based Authentication (used in this Lab)
- 3) Digest Authentication
- 4) Client Certificate Authentication (*)



11) Add the method in the controller to handle request for /login

```
@GetMapping("/login")
public String loginPage(...) {
    ....
    //Get the Cart Data if any and store in Session
    //Get the Last Accessed Time from DB.

    return "mylogin";
}
```

A. <http://localhost:56789/login>

No Parameters going to server with names error or logout
** Nothing display in login Page

B. <http://localhost:56789/login?error>

error Parameter going to server

```
<c:if test="${param.error != null}">
    Invalid username and password.
</c:if>
```

C. <http://localhost:56789/login?logout>

logout Parameter going to server

```
<c:if test="${param.logout != null}">
    You have logged out successfully.
</c:if>
```

Steps for Task 4: Customize Logout Page

12) Update HttpSecurity with logout configuration.

```
.logout()
.invalidateHttpSession(true)
.logoutSuccessUrl("/login?logout")
```

13) Add the method in the controller to handle request for /logout

```
@GetMapping("/logout")
public String logoutPage(.....) {
    //Store the Cart Data if any into the database
    Tasks At Logout time
}
```



Steps for Task 5: Use the JDBC Authentication.

14) Add the Maven dependency for spring data jpa

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```

15) Add the Maven dependency for mysql

```
<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
</dependency>
```

16) Update application.properties

17) Setup database tables

```
create database myjlcdb;
use myjlcdb;
```

```
CREATE TABLE myusers (
    username VARCHAR(15) PRIMARY KEY,
    password VARCHAR(200) NOT NULL ,
    firstname VARCHAR(15) ,
    lastname VARCHAR(15) ,
    email VARCHAR(35) ,
    phone VARCHAR(10) ,
    active TINYINT NOT NULL
);
```

```
CREATE TABLE myroles (
    user_role_id int PRIMARY KEY,
    myusername varchar(15) NOT NULL,
    role varchar(25) NOT NULL,
    FOREIGN KEY (myusername) REFERENCES myusers(username)
);
```

```
insert into myusers values('myhello','myhello','hello','hello','hello@jlc',12345,1);
```

```
insert into myroles values(1,'myhello','ROLE_CUSTOMER');
```



18) Write SQL queries for fetching Users and User Roles in configureGlobal() method.

19) Update AuthenticationManager with JdbcAuthentication

```
auth.jdbcAuthentication()  
    .dataSource(dataSource)  
    .usersByUsernameQuery(usersQuery)  
    .authoritiesByUsernameQuery(rolesQuery);
```

20) Configure the PasswordEncoder

```
@Bean  
public BCryptPasswordEncoder passwordEncoder() {  
    return new BCryptPasswordEncoder();  
}
```

Steps for Task 6: Implement Registration

21) Design register.jsp

22) Update mymenu.jsp

```
| <a href='<c:url value="showRegister"/>'>Register</a>
```

23) Write User.java

24) Update the UserController

```
@GetMapping("/showRegister")  
public String showRegisterPage(Model model) {  
    ....  
    return "register";  
}  
  
@PostMapping("/registerMyUser")  
public String registerUser(@ModelAttribute("myuser") User user, BindingResult  
bindingResult,Model model) {  
    ....  
    return "index";  
}
```

25) Write UserService and UserServiceImpl

26) Write UserDAO and UserDAOImpl

27) Deploy to Server and open the App URL

28) Do some 6 registrations using register.jsp

srinivas , vas , sd , ds , sri ,hello

29) Insert user related myroles data in db.

```
INSERT INTO myroles VALUES (1,'srinivas', 'ROLE_ADMIN');
INSERT INTO myroles VALUES (2,'vas', 'ROLE_STOREKEEPER');
INSERT INTO myroles VALUES (3,'sd', 'ROLE_CUSTOMER');
INSERT INTO myroles VALUES (4,'ds', 'ROLE_CUSTOMER');
INSERT INTO myroles VALUES (5,'sri', 'ROLE_CUSTOMER');
INSERT INTO myroles VALUES (6,'sri', 'ROLE_ADMIN');
INSERT INTO myroles VALUES (7,'hello', 'ROLE_CUSTOMER');
```

30) Disable user whose name is hello.

```
update myusers set active=0 where username='hello';
```

31) Deploy and Check your Luck.

Steps for Task 7: Display the Links according to Role.

32) Update index.jsp

```
<sec:authorize access="hasRole('ADMIN')">
<a href='<c:url value="deleteBook"/>' > Delete Book </a>
</sec:authorize>

<sec:authorize access="hasRole('CUSTOMER')">
<a href='<c:url value="placeOrder"/>' > Place Order </a>
</sec:authorize>
```

Lab30: Files Required

16. index.jsp	Updated in Lab30
17. myheader.jsp	Same as Lab28
18. myfooter.jsp	Same as Lab28
19. mymenu.jsp	Updated in Lab30
20. viewBooks.jsp	Same as Lab28
21. addBook.jsp	Same as Lab28
22. editBook.jsp	Same as Lab28
23. deleteBook.jsp	Same as Lab28
24. placeOrder.jsp	Same as Lab28
25. invalidAccess.jsp	Same as Lab28
26. mylogin.jsp	Newly Added in Lab30
27. register.jsp	Newly Added in Lab30
28. UserController.java	Updated in Lab30
29. User.java	Newly Added in Lab30
30. UserService.java	Newly Added in Lab30
31. UserServiceImpl.java	Newly Added in Lab30
32. UserDAO.java	Newly Added in Lab30
33. UserDAOImpl.java	Newly Added in Lab30
34. JLCWebConfig.java	Updated in Lab30
35. StartMyBootApplication.java	Same as Lab15
36. application.properties	Updated in Lab30
37. pom.xml	Updated in Lab30

16. index.jsp

```

<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<%@ taglib prefix="sec" uri="http://www.springframework.org/security/tags" %>

<html>
<head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>

```



```
<div class="card">
  <div class="card-body">
    <c:import url="mymenu.jsp"/>
    <div class="container" align="center">
      <h2>

      <a href='<c:url value="viewBooks"/>'> View Books</a>

      <sec:authorize access="hasRole('ADMIN') or hasRole('STOREKEEPER')">
        <br/><br/>
        <a href="addBook"> Add Book</a>
        <br/><br/>
        <a href="editBook"> Edit Book</a>
      </sec:authorize>

      <sec:authorize access="hasRole('ADMIN')">
        <br/><br/>
        <a href="deleteBook"> Delete Book</a>
      </sec:authorize>

      <sec:authorize access="hasRole('CUSTOMER')">
        <br/><br/>
        <a href="placeOrder"> Place Order</a>
      </sec:authorize>
    </h2>

  </div>
</div>
<c:import url="myfooter.jsp"/>
</body>
</html>
```

17. myheader.jsp

18. myfooter.jsp

19. mymenu.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<%@ taglib prefix="sec" uri="http://www.springframework.org/security/tags" %>
```

```
<div class="container" align="right">
```

```
<h2>
```

```
<a href="/">Home</a>
```



| Welcome : \${pageContext.request.userPrincipal.name} \${MyRoles}

```
<sec:authorize access="! isAuthenticated()">
| <a href="login">Login</a>
| <a href="showRegister">Register</a>
</sec:authorize>
```

```
<sec:authorize access="isAuthenticated()">
<a href="logout"> Logout</a>
</sec:authorize>
```

```
</h2>
</div>
```

- 20. [viewBooks.jsp](#)
- 21. [addBook.jsp](#)
- 22. [editBook.jsp](#)
- 23. [deleteBook.jsp](#)
- 24. [placeOrder.jsp](#)
- 25. [invalidAccess.jsp](#)
- 26. [mylogin.jsp](#)

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
```

```
<html> <head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<div class="card">
  <div class="card-body" align="center">
    <div class="container-md">
      <h2 align="center">
        <c:if test="${param.error != null}">
          <font color="red" size=5> Invalid username and password. </font>
        </c:if>
        <c:if test="${param.logout != null}">
          <font color="red" size=5>You have logged out successfully.</font>
        </c:if>
      </h2>
    </div>
  </div>
</div>
```



```
<div class="container form-group">
<form name='loginForm' action="<c:url value='/login' />" method='POST'>
<table class="table">
<tr>
<td align="center" colspan="2">
<h2>User Login Form</h2>
</td> </tr>
<tr>
<td><label for="myusername"> Username </label></td>
<td><input type='text' name='myusername' id="myusername" class="form-control form-
control-lg">
</td> </tr>
<tr>
<td>Password</td>
<td><input type='password' name='mypassword' class="form-control form-control-lg" />
</td></tr>
<tr>
<td align="center" colspan="2">
<input type=hidden name="_csrf.parameterName" value="{_csrf.token}" />
<input type="submit" value="Login Now" class="btn btn-primary btn-lg">
</td> </tr>
</table>
</form>
</div>
<div class="container">
<h2> <b> <a href='<c:url value="showRegister" />'> New User Register Here </a>
</b></h2>
</div>
</div> </div>
<c:import url="myfooter.jsp"/>
</body> </html>
```

27. register.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<%@ taglib prefix="form" uri="http://www.springframework.org/tags/form"%>

<html> <head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
```



```
<div class="card">
  <div class="card-body" align="center">
    <hr>
  <div class="container">
    <h2 align="center">
      <c:if test="${param.error != null}">
        <font color="red" size=5> Invalid username and password. </font>
      </c:if>

      <c:if test="${param.logout != null}">
        <font color="red" size=5>You have logged out successfully.</font>
      </c:if>
    </h2>
  </div>
</div>
```

```
<div class="container">
<form:form action="registerMyUser" method="post" modelAttribute="myuser">
<table class="table">
<tr>
<td align="center" colspan="2">
<h2>User Registration Form</h2>
</td> </tr>
<tr>
<td>Username</td>
<td><form:input path="username" class="form-control form-control-lg"/></td>
<td> <form:errors path="username" /></td>
</tr>
<tr>
<td>Password</td>
<td><form:password path="password" class="form-control form-control-lg"/></td>
<td> <form:errors path="password" /></td>
</tr>
<tr>
<td>First Name</td>
<td><form:input path="firstname" class="form-control form-control-lg"/></td>
<td> <form:errors path="firstname" /></td>
</tr>
<tr>
<td>Las Name</td>
<td><form:input path="lastname" class="form-control form-control-lg"/></td>
<td> <form:errors path="lastname" /></td>
</tr>
<tr>
<td>Email</td>
```

```

<td><form:input path="email" class="form-control form-control-lg"/></td>
<td> <form:errors path="email" /></td>
</tr>
<tr>
<td>Phone</td>
<td><form:input path="phone" class="form-control form-control-lg"/></td>
<td> <form:errors path="phone" /></td>
</tr>
<tr>
<td align="center" colspan="2">
<input type="hidden" name="${_csrf.parameterName}" value="${_csrf.token}" />
<input type="submit" value=" Register Bow" class="btn btn-primary btn-lg">
</td>
</tr>
</table>
</form:form>
</div>
<div class="container">
<h2>   <b> <a href='<c:url value="login"/>'> Already Registered Login Here </a> </b></h2>
</div>
</div>
</div>
<c:import url="myfooter.jsp"/>
</body>  </html>

```

28. UserController.java

```
package com.coursecube.springboot;
```

```

import java.util.*;
import javax.servlet.http.*;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.context.annotation.Bean;
import org.springframework.security.core.*;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.web.authentication.logout.*;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.*;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */

```



```
@Controller
@SessionAttributes("MyRoles")
public class UserController {

    @Autowired
    UserService userService;

    @GetMapping( "/" )
    public String showIndexPage(HttpSession session) {
        System.out.println("** showIndexPage **");

        List<String> myroles=getUserRoles();
        System.out.println(myroles);
        session.setAttribute("MyRoles", myroles);

        return "index";
    }

    @GetMapping( "/viewBooks" )
    public String showViewBooksPage(HttpSession session) {
        System.out.println("** showViewBooksPage **");

        List<String> myroles=getUserRoles();
        System.out.println(myroles);
        session.setAttribute("MyRoles", myroles);

        return "viewBooks";
    }

    @GetMapping( "/addBook" )
    public String showAddBookPage(HttpSession session) {
        System.out.println("** showAddBookPage **");

        List<String> myroles=getUserRoles();
        System.out.println(myroles);
        session.setAttribute("MyRoles", myroles);

        return "addBook";
    }

    @GetMapping( "/editBook" )
    public String showEditBookPage(HttpSession session) {
        System.out.println("** showEditBookPage **");
        List<String> myroles=getUserRoles();
```



```
System.out.println(myroles);  
session.setAttribute("MyRoles", myroles);
```

```
return "editBook";  
}
```

```
@GetMapping( "/deleteBook" )  
public String showDeleteBookPage(HttpSession session) {  
    System.out.println("** showDeleteBookPage **");
```

```
    List<String> myroles=getUserRoles();  
    System.out.println(myroles);  
    session.setAttribute("MyRoles", myroles);
```

```
    return "deleteBook";  
}
```

```
@GetMapping( "/placeOrder" )  
public String showPlaceOrderPage(HttpSession session) {  
    System.out.println("** showPlaceOrderPage **");
```

```
    List<String> myroles=getUserRoles();  
    System.out.println(myroles);  
    session.setAttribute("MyRoles", myroles);
```

```
    return "placeOrder";  
}
```

```
@GetMapping("/showRegister")  
public String showRegisterPage(Model model) {  
    System.out.println("** showRegisterPage **");  
    User myuser=new User();  
    model.addAttribute("myuser",myuser);  
    return "register";  
}
```

```
@PostMapping("/registerMyUser")  
public String registerUser(@ModelAttribute("myuser") User user, BindingResult  
bindingResult,Model model) {  
    System.out.println("** registerUser **");  
    //Do Validations  
    userService.registerUser(user);  
    return "index";  
}  
//Spring Security see this :
```



```
@GetMapping("/login")
public String loginPage(
    @RequestParam(value = "error", required = false) String error,
    @RequestParam(value = "logout", required = false) String logout, HttpSession session) {
    System.out.println("** loginPage **");

    List<String> myroles=getUserRoles();
    System.out.println(myroles);
    session.setAttribute("MyRoles", myroles);

    return "mylogin";

}

@GetMapping("/logout")
public String logoutPage(HttpServletRequest request, HttpServletResponse response) {
    System.out.println("** logoutPage **");

    List<String> myroles = getUserRoles();
    System.out.println(myroles);

    Authentication auth = SecurityContextHolder.getContext().getAuthentication();
    if (auth != null) {
        new SecurityContextLogoutHandler().logout(request, response, auth);
    }
    return "redirect:/login?logout";
}

private List<String> getUserRoles() {
    Authentication auth = SecurityContextHolder.getContext().getAuthentication();
    List<String> myroles = new ArrayList<String>();
    if (auth != null) {
        Collection<GrantedAuthority> col = (Collection<GrantedAuthority>) auth.getAuthorities();
        for (GrantedAuthority gauth : col)
            myroles.add(gauth.getAuthority());
    }
    return myroles;
}
}
```

29. User.java



```
package com.coursecube.springboot;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
public class User {
    private String username;
    private String password;
    private String firstname;
    private String lastname;
    private String email;
    private String phone;
    private int active;

    //Setters and Getters

}
```

30. **UserService.java**

```
package com.coursecube.springboot;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
public interface UserService {
    public void registerUser(User user) ;
}
```

31. **UserServiceImpl.java**

```
package com.coursecube.springboot;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
import org.springframework.stereotype.Service;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
@Service
public class UserServiceImpl implements UserService {

    @Autowired
```



```
private BCryptPasswordEncoder bCryptPasswordEncoder;

@Autowired
UserDAO userDAO;

public void registerUser(User user) {
    System.out.println(user.getPassword());
    user.setPassword(bCryptPasswordEncoder.encode(user.getPassword()));
    System.out.println(user.getPassword());
    user.setActive(1);

    userDAO.registerUser(user);
}
}
```

32. UserDAO.java

```
package com.coursecube.springboot;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
public interface UserDAO {
    public void registerUser(User user) ;
}
```

33. UserDAOImpl.java

```
package com.coursecube.springboot;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.jdbc.core.JdbcTemplate;
import org.springframework.stereotype.Repository;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
@Repository
public class UserDAOImpl implements UserDAO {

    @Autowired
    JdbcTemplate jdbcTemp;

    @Override
```



```
public void registerUser(User user) {  
String sql="insert into myusers values(?,?,?,?,?,?)";  
jdbcTemp.update(sql,user.getUsername(),user.getPassword(),user.getFirstname(),user.getLas  
tname(),user.getEmail(),user.getPhone(),user.getActive());  
}  
}
```

34. JLCWebConfig.java

```
package com.coursecube.springboot;
```

```
import javax.sql.DataSource;  
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.context.annotation.*;  
import org.springframework.security.config.annotation.authentication.builders.*;  
import org.springframework.security.config.annotation.web.builders.HttpSecurity;  
import org.springframework.security.config.annotation.web.configuration.*;  
import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;  
import org.springframework.web.servlet.config.annotation.*;  
import org.springframework.web.servlet.view.*;  
/*  
 * @Author : Srinivas Dande  
 * @Company : CourseCube Private Limited  
 * @Website : www.coursecube.com  
 */  
@EnableWebMvc  
@Configuration  
@ComponentScan({ "com.coursecube.springboot" })  
public class JLCWebConfig extends WebSecurityConfigurerAdapter implements  
WebMvcConfigurer {  
  
@Autowired  
DataSource dataSource;  
  
@Bean  
public InternalResourceViewResolver viewResolver() {  
InternalResourceViewResolver viewResolver = new InternalResourceViewResolver();  
viewResolver.setViewClass(JstlView.class);  
viewResolver.setPrefix("/WEB-INF/myjsps/");  
viewResolver.setSuffix(".jsp");  
return viewResolver;  
}  
  
@Override
```



```
public void addResourceHandlers(ResourceHandlerRegistry registry) {  
    registry.addResourceHandler("/webjars/**").addResourceLocations("classpath:/META-INF/resources/webjars/");  
}
```

@Autowired

```
public void configureGlobal(AuthenticationManagerBuilder auth) throws Exception {
```

```
    String usersQuery="select username,password, active from myusers where username=?";  
    String rolesQuery="select username, role from myroles where username=?";
```

```
    auth.jdbcAuthentication()  
        .dataSource(dataSource)  
        .usersByUsernameQuery(usersQuery)  
        .authoritiesByUsernameQuery(rolesQuery);
```

```
}
```

@Override

```
protected void configure(HttpSecurity http) throws Exception {
```

```
    http.authorizeRequests()  
        .antMatchers("/deleteBook**").access("hasRole('ROLE_ADMIN')")  
        .antMatchers("/addBook**").access("hasAnyRole('ROLE_ADMIN','ROLE_STOREKEEPER')")  
        .antMatchers("/editBook**").access("hasAnyRole('ROLE_ADMIN','ROLE_STOREKEEPER')")  
        .antMatchers("/placeOrder**").access("hasAnyRole('ROLE_CUSTOMER')")  
        .and()  
        .formLogin()  
        .loginPage("/login")  
        .failureUrl("/login?error")  
        .usernameParameter("myusername").passwordParameter("mypassword")  
        .and()  
        .logout().invalidateHttpSession(true)  
        .logoutSuccessUrl("/login?logout")  
        .and()  
        .exceptionHandling().accessDeniedPage("/WEB-INF/myjsps/invalidAccess.jsp")  
        .and().csrf().and()  
        .sessionManagement()  
        .enableSessionUrlRewriting(true)  
        .maximumSessions(1);  
}
```

@Bean

```
public BCryptPasswordEncoder passwordEncoder() {  
    BCryptPasswordEncoder bCryptPasswordEncoder = new BCryptPasswordEncoder();
```



```
return bCryptPasswordEncoder;  
}  
}
```

35. StartMyBootApplication.java

36. application.properties

```
server.port=56789
```

```
spring.datasource.driver-class-name=com.mysql.jdbc.Driver  
spring.datasource.url=jdbc:mysql://localhost:3306/myjldb  
spring.datasource.username=root  
spring.datasource.password=srinivas
```

```
spring.datasource.hikari.connectionTimeout=20000  
spring.datasource.hikari.maximumPoolSize=5
```

```
logging.level.root=INFO  
logging.pattern.console=%-5level %logger{36} - %msg%n
```

```
spring.jpa.show-sql=true  
spring.jpa.hibernate.ddl-auto= update
```

37. pom.xml

```
<project...>  
<modelVersion>4.0.0</modelVersion>  
  
<groupId>com.coursecube.springboot</groupId>  
<artifactId>Lab30</artifactId>  
  
<version>1.0</version>  
<packaging>jar</packaging>  
<name>MyLab30</name>  
<url>http://maven.apache.org</url>  
  
<properties>  
<java.version>1.8</java.version>  
</properties>  
  
<parent>  
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-starter-parent</artifactId>
```



```
<version>2.1.9.RELEASE</version>  
</parent>
```

```
<dependencies>  
<dependency>  
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-starter-web</artifactId>  
</dependency>
```

```
<dependency>  
<groupId>org.apache.tomcat.embed</groupId>  
<artifactId>tomcat-embed-jasper</artifactId>  
</dependency>
```

```
<dependency>  
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-starter-security</artifactId>  
</dependency>
```

```
<dependency>  
<groupId>org.springframework.security</groupId>  
<artifactId>spring-security-taglibs</artifactId>  
</dependency>
```

```
<dependency>  
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-starter-data-jpa</artifactId>  
</dependency>
```

```
<dependency>  
<groupId>mysql</groupId>  
<artifactId>mysql-connector-java</artifactId>  
<version>5.1.6</version>  
</dependency>
```

```
<dependency>  
<groupId>jstl</groupId>  
<artifactId>jstl</artifactId>  
<version>1.2</version>  
</dependency>
```



```
<dependency>
<groupId>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>
<build>
<finalName>MyLab30</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```